Math 698

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In these notes we only give the key definitions, lemmas and theorems as well as the sketch of proof. The details can be found in the appended handwritten notes.

Our main goal is the following theorem: $\overline{\overset{\circ}{L_n^d}} = L_n^d$. Since we've already proved $\overline{\overset{\circ}{L_n^d}} \supseteq L_n^d$ previously, it suffices to prove that L_n^d is closed. Let f_k be a sequence of polynomials in L_n^d converging to f, then we need to prove that $f \in L_n^d$, i.e.

1. $\mathrm{supp}(f)$ is M-convex;
2. $\partial_i f \in L_n^{d-1}$.

We use induction. Suppose this holds for $d-1$, then the second part holds because

$$\partial_i f = \partial_i(\lim f_k) = \lim(\partial_i f_k) \in L_n^{d-1}.$$

The second equality holds since H_n^d is a finite dimensional linear space and ∂_i is a linear operator. Therefore, it suffices to prove that $\mathrm{supp}(f)$ is M-convex.

Definition (c-Rayleigh)**.** *We say that f is c-Rayleigh if f has nonnegative coefficients and*

$$\partial^\alpha f(w)\partial^{\alpha+e_i+e_j} f(w) \le c\partial^{\alpha+e_i} f(w)\partial^{\alpha+e_j} f(w) \text{ for all } i,j \in [n], \alpha \in \mathbb{N}^n, w \in \mathbb{R}^n_{\ge 0}.$$

Note that c-Rayleigh is closed under limits and the following operations.

Proposition 1. *If f is c-Rayleigh, then the following*

- $\partial_i f$;
- $f\backslash i$, *the polynomial obtained from f by evaluating $w_i = 0$;*
- $f(a_1 w_1, \cdots, a_n w_n)$ *for* $(a_1, \cdots, a_n) \in \mathbb{R}^n_{\ge 0}$;
- $f(a_1 + w_1, \cdots, a_n + w_n)$ *for* $(a_1, \cdots, a_n) \in \mathbb{R}^n_{\ge 0}$.

are c-Rayleigh.

Lemma 1. *Suppose $f \in L_n^d$ nonzero and $d \ge 2$, then the Hessian matrix $\mathcal{H}_f(w)$ has exactly one positive eigenvalue for all $w \in \mathbb{R}^n_{>0}$.*

Proof. (sketch) $\mathcal{H}_f(w)$ has at least one positive eigenvalue since $f \neq 0$ has nonnegative coefficients. Therefore, it suffices to prove that $\mathcal{H}_f(w)$ has at most one positive eigenvalue. Recall the homotopy $T_n(\theta, S(\theta, f))$ we defined previously (in Marcus talk) that deforms any $f \in \overset{\circ}{L_n^d}$ to $T_n(1, (w_1 + \cdots + w_n)^d)$. Therefore, the whole space $\overset{\circ}{L_n^d}$ is connected. As a consequence, it suffices to prove that there exists $f \in L_n^d$ such that $\mathcal{H}_f(w)$ has exactly one positive eigenvalue, and that if $f \in \overset{\circ}{L_n^d}$ then $\mathcal{H}_f(w)$ is nonsingular. The former one is equivalent to $P(z) := z^T \mathcal{H}_f(w) z = \sum z_i z_j \partial_i \partial_j f(w)$ is stable; the latter one is proved by induction and the fact that $\ker \mathcal{H}_f(w) = \bigcap \ker \mathcal{H}_{\partial_i f}(w)$ for all $w \in \mathbb{R}^n_{>0}$ if $\mathcal{H}_{\partial_i f}(w)$ has exactly one positive eigenvalue for all $w \in \mathbb{R}^n_{>0}$. □

Theorem 1. *If $f \in L_n^d$, then f is $2(1-\frac{1}{d})$-Rayleigh.*

Proof. (sketch) By considering the case $d = 2$, we can find that $\mathcal{H}_f(w)$ has exactly one positive eigenvalue implies that f is $2(1-\frac{1}{d})$-Rayleigh. Lemma 1 completes the proof. □

Motivated by the independent sets of matroids, we give the following definition.

Definition ($M^\#$-convex set)**.** *A subset $J^\# \subset \mathbb{N}^n$ is $M^\#$-convex if it satisfies the following conditions:*

① $0 \in J^\#$

② *If* $\beta \in J^\#, \alpha \leq \beta$, *then* $\alpha \in J^\#$ *(where the partial order* $\alpha \leq \beta$ *is defined as* $\alpha_i \leq \beta_i$ *for all* i*)*

③ *If* $\alpha, \beta \in J^\#$ *and* $|\alpha|_1 < |\beta|_1$, *then there exists* i *such that* $\alpha_i < \beta_i$ *and* $\alpha + e_i \in J^\#$

The following proposition shows that the relation between M-convex set and $M^\#$-convex set is exactly like the relation between basis and independent sets of matroids.

Proposition 2. *For finite* $J^\# \subset \mathbb{N}^n$, *let* $m := \max_{\alpha \in J^\#}\{|\alpha|_1\}$, *and denote* $J := \{\alpha \in J^\# \mid |\alpha|_1 = m\}$. *Then* J *is* M*-convex whenever* $J^\#$ *is* $M^\#$*-convex.*

Proof. Routine check. □

Lemma 2. *Let* f *be a* c*-Rayleigh polynomial. If* $supp(f)$ *satisfies the 1st condition of the definition of* $M^\#$*-convex set, then it also satisfies the 2nd and 3rd condition, i.e., if* $f(0) \neq 0$ *then* $supp(f)$ *is* $M^\#$*-convex.*

Proof. ① ⇒ ②: Suppose that the polynomial f and $\alpha \leq \beta$ constitute a minimal counterexample to ② with respect to the degree d, number of variables n, α and $|\beta|_1$, i.e., f is c-Rayleigh, $f(0) \neq 0$, $J^\# := \text{supp}(f)$, $\beta \in J^\#$, $\alpha \notin J^\#$, and $(d, n, \alpha, |\beta|_1)$ is minimal among these counterexamples. We can prove the following propositions for the counterexample:

- For all j, $\beta_j \neq 0$: otherwise, $(f\backslash j, \alpha \leq \beta)$ gives a smaller counterexample.
- $\alpha = e_i$ for some i: suppose not, then because $\alpha \notin J^\#$ and $0 \in J^\#$, we have $\alpha \neq 0$, which implies that there exists i such that $\alpha_i > 0$. If $e_i \in J^\#$ then $(\partial_i f, \alpha - e_i \leq \beta - e_i)$ gives a smaller counterexample; otherwise $e_i \notin J^\#$, then $(f, e_i \leq \beta)$ gives a smaller counterexample.
- For all j, $e_j \notin J^\#$: If $e_j \in J^\#$, then $e_i + e_j \in J^\#$ (note that $\alpha = e_i$ as previously proved) since otherwise $(\partial_j f, e_i \leq \beta - e_j)$ gives a smaller counterexample. As a result, $f(0) \neq 0$, $\partial_i \partial_j f(0) \neq 0$ and $\partial_i f(0) = 0$, which implies that f is not c-Rayleigh, a contradiction.
- For all $0 \neq \gamma \in J^\#$, $|\gamma|_1 \geq |\beta|_1$: otherwise, $(f, e_j \leq \gamma)$ gives a smaller counterexample for some j such that $\gamma_j > 0$.

Now choose l, k such that $e_l + e_k \leq \beta$ and $w = (\epsilon, \cdots, \epsilon)$ for some $\epsilon > 0$. By the minimality of $|\beta|_1$, the lowest degree with respect to ϵ of $f\partial_l\partial_k f(w)$ is $|\beta|_1 - 2$, while the lowest degree with respect to ϵ of $\partial_l f(w)\partial_k f(w)$ is $2|\beta|_1 - 2$, which implies that f is not c-Rayleigh, a contradiction.

①② ⇒ ③: Suppose that the polynomial f and $\alpha, \beta \in J^\# := \text{supp}(f)$ constitute a minimal counterexample to ③ with respect to the degree d, number of variable n and $|\alpha|_1$, i.e., f is c-Rayleigh, $f(0) \neq 0$, $J^\#$ satisfies condition ② and for all j, $\alpha + e_j \notin J^\#$ whenever $\alpha_j < \beta_j$, and $(d, n, |\alpha|_1)$ is minimal among these counterexamples. For any $\gamma \in \mathbb{N}^n$, define $S(\gamma) := \{i \mid \gamma_i > 0\}$. We can prove the following propositions for the counterexample:

- $S(\alpha) \cap S(\beta) = \emptyset$: otherwise $(\partial_j f, \alpha - e_j, \beta - e_j)$ gives a smaller counterexample for $j \in S(\alpha) \cap S(\beta)$.
- $S(\alpha) \cup S(\beta) = [n]$: otherwise $(f\backslash j, \alpha, \beta)$ gives a smaller counterexample for $j \notin S(\alpha) \cup S(\beta)$.

By ②, $\alpha \neq 0$, so there exists $i \in S(\alpha)$, and $\alpha - e_i \in \text{supp}(f)$. Since $|\alpha|_1$ is minimal, $(\alpha - e_i, \beta)$ is not a counterexample, implying $\exists j$ such that $(\alpha - e_i)_j < \beta_j$ and $\alpha - e_i + e_j \in J^\#$. Since $j \in S(\alpha - e_i + e_j) \cap S(\beta)$, $(\alpha - e_i + e_j, \beta)$ must not be a counterexample (otherwise $(\partial_j f, \alpha - e_i, \beta - e_j)$ gives a counterexample), meaning $\exists k$ such that $(\alpha - e_i + e_j)_k < \beta_k$ and $\alpha - e_i + e_j + e_k \in \text{supp}(f)$. Note also that since (α, β) fails the augmentation property, we must have $\alpha + e_j \notin \text{supp}(f)$ and $\alpha + e_k \notin \text{supp}(f)$.

Now consider $g := \partial^{\alpha - e_i} f$ evaluated at $w = (1, \ldots, 1, w_i, 1, \ldots, 1)$. The term $g(w)\partial_j\partial_k g(w)$ is strictly increasing with w_i because $\alpha, \alpha - e_i + e_j + e_k \in \text{supp}(f)$. However, $c\partial_j g(w)\partial_k g(w)$ is independent of w_i since $\alpha + e_j \notin \text{supp}(f)$, $\alpha + e_k \notin \text{supp}(f)$ and ② imply that $\partial_j g$ and $\partial_k g$ have no w_i terms. This yields a contradiction, completing the proof. □

Theorem 2. *If* $0 \neq f \in H_n^d$ *is* c*-Rayleigh, then* $supp(f)$ *is* M*-convex.*

Proof.

$$
\begin{aligned}
&f \text{ is } c\text{-Rayleigh} \\
\Rightarrow &g(w_1, \cdots, w_n) := f(w_1 + 1, \cdots, w_n + 1) \text{ is } c\text{-Rayleigh} \\
\Rightarrow &\text{supp}(g) \text{ is } M^\#\text{-convex} \\
\Rightarrow &\text{supp}(f) \text{ is } M\text{-convex.}
\end{aligned}
$$

□

Theorem 3. *L_n^d is closed.*

Proof. Let f_k be a sequence of polynomials in L_n^d converging to f. As stated at the beginning of this note, it suffices to prove that $\mathrm{supp}(f)$ is M-convex. By Theorem 1, f_k are $2(1-\frac{1}{d})$-Rayleigh. Since c-Rayleigh is closed under limits, f is $2(1-\frac{1}{d})$-Rayleigh. By Theorem 2, $\mathrm{supp}(f)$ is M-convex. □

Week 5

$\overline{\overset{\circ}{L}{}_n^d} = L_n^d$

(2,13) $\overline{\overset{\circ}{L}{}_n^d} \supseteq L_n^d$

Goal: $\overline{\overset{\circ}{L}{}_n^d} \subseteq L_n^d$

Have: $\overset{\circ}{L}{}_n^d \subseteq L_n^d$ (since $\mathrm{coeff} > 0$, M-convex)

L_n^d: M-convex

Goal: L_n^d closed

$\{f_k\} \subset L_n^d$, $f_k \to f \Rightarrow f \in L_n^d$

① $\mathrm{supp}(f)$ M-convex Goal

② $\partial_i f \in L_n^{d-1}$

$\partial_i(\lim f_k) = \lim(\partial_i f_k)$, $\partial_i f_k \in L_n^{d-1}$

∂_i 作用第 i 有限值 $\Rightarrow$ 连续

base case: $L_n^2 = S_n^2$ closed (Hurwitz's thm. P7)

$of \in H_n^d$, $d \geq 2$, $c_\alpha \geq 0$

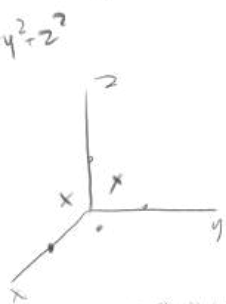
C-Reyleigh: nonnegative coeff $\forall i,j, \alpha, w \in \mathbb{R}^n_{\geq 0}$

$$\partial^\alpha f(w)\, \partial^{\alpha+e_i+e_j} f(w) \leq c\, \partial^{\alpha+e_i} f(w)\, \partial^{\alpha+e_j} f(w)$$

$f_k \in L_n^d \ni f_k$ C-Reyleigh $\Rightarrow f$ C-Reyleigh $\ni f \in M_n^d$

Q. why C-Reyleigh $\propto$ M-convex ? 某个 w 不好到他大一点不行

$x^2+xy+y^2+z^2$



∂_x:



LHS: F(z)

RHS: No z

∂_y



Goal: Lorentz $\Rightarrow$ C-Reyleigh

Suppose $\alpha = 0$ (Idea: $\partial^\alpha f \in L_n^d$, finite α, $\max_\alpha c$)

Goal: $f(w)\, \partial_i \partial_j f(w) \leq c\, \partial_i f(w)\, \partial_j f(w)$

Rmk $df(w) = \sum w_i \partial_i f(w)$ pf $\sum w_i \partial_i w^\alpha = (\sum \alpha_i) w^\alpha$

$$\frac{1}{d(d-1)} (w^T H w)(e_i^T H e_j) \leq c \frac{1}{(d-1)^2} (e_i^T H w)(w^T H e_j)$$

Case $d=2$:

$$\frac{1}{2}(w^T H w)(e_i^T H e_j) \le c\,(e_i^T H w)(w^T H e_j)$$

H one positive eigenvalue.

(By Cauchy's interlacing thm)

H one >0 eigen on $\mathrm{span}\{w, v_t := e_i + t e_j\}$

$$(w^T H w)(v_t^T H v_t) - (w^T H v_t)^2 \le 0$$

$$(w^T H w)(e_i^T H e_i) + t\,2(w^T H w)e_i^T H e_j + t^2 (w^T H w)(e_j^T H e_j)$$

$$-(w^T H e_i + t\,w^T H e_j)^2 \le 0$$

$$t^2 (w^T H e_j)^2 + t\left[\underbrace{2(w^T H e_i)}_{X}\underbrace{(w^T H e_j)}_{Y} - 2(w^T H w)(e_i^T H e_j)\right] + (w^T H e_i)^2 \ge 0, \ \forall t$$

$\Rightarrow \frac{1}{4}\Delta = (X-Y)^2 - X^2 = Y^2 - 2XY \le 0 \Rightarrow Y \le 2X \Rightarrow c = 2$

key: H one positive eigenvalue $\Rightarrow c = 2\left(\frac{d-1}{d}\right)$

Fix $w \in \mathbb{R}^n_{>0}$ Goal: $\forall d$, $H_f(w)$ one >0 eigen

thm. $\forall f \in L^d_w$, $H_f(w)$ exactly one >0 eigen.

(at least one >0, since 无零生拒)

suffice. $\forall f \in \mathring{L}^d_w$ ——

Recall $T_n(\theta, S(\theta, f))$ homotopy

$\theta = 0$ f $\Rightarrow \mathring{L}^d_w$ connected

$\theta = 1$ $T_n(1, (w_1 + \cdots + w_n)^d)$

Target:

① Find one $f \in \mathring{L}^d_n$ exactly one >0 eigen

($\Leftrightarrow z^T H_f(w) z = \sum z_i z_j \partial_i \partial_j f(w)$ stable)

② $f \in \mathring{L}^d_n \Rightarrow H_f(w)$ nonsingular

① f strictly stable $\Rightarrow g(z) = f(z+w)$ stable

Taylor. quadratic part

$\Rightarrow f_1 = f(z + (x, x, \cdots, x))$ $\sim$次 $\cdot x^d + \sim$次 $\cdot x^{d-1} + \cdots + d$次

$f_2 = \partial_x^{d-2} f_1$ ~ 0次 $\cdot x^2 + \sim 1$次 $\cdot x + \sim 2$次

$f_3 = z_1^d \cdots z_n^d x^2 f_2(z_1^{-1} \cdots z_n^{-1} \cdot x^{-1})$ $\sim \widetilde{0次} + \sim \widetilde{1次} x + \sim \widetilde{2次} x^2$

$f_4 = \partial_x^2 f_3$ $\sim \widetilde{2次}$

$f_5 = z_1^d \cdots z_n^d f_4(z_1^{-1} \cdots z_n^{-1})$ 2次

② Induction: Suppose $f \in L_n^{d-1} \Rightarrow H_f(w)$ nonsingular.

then $f \in L_n^{d-1} \Rightarrow H_f(w)$ one ≥ 0 eigen

Now fix $f \in L_n^d$

Claim: $\ker H_f(w) \subseteq \bigcap_{i=1}^{n} \ker H_{\partial_i f}(w)$ Suffice

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Given $H_f(w) z = 0$

(Note $z^T H_{\partial_i f} z \leq 0$ (Why? $\nabla f \in$ span $\{w, \ldots\} \perp z$)
$(= 0$ iff $H_{\partial_i f} z = 0)$

$$\partial_i \partial_j f = (d-2)\, \partial_i \partial_j f(w) = \sum_i w_i \partial_i \partial_i \partial_j f(w)$$
$$= w^T (\partial_i \partial_j \partial f)_i$$
$$(d-2)\, e_i^T H_f = w^T H_{\partial_i f}$$
$$\Rightarrow w^T H_{\partial_i f} z = 0, \ \forall i$$

$$(d-2) H_f = \sum w_i H_{\partial_i f}$$
$$(d-2) z^T H_f z = \sum w_i (z^T H_{\partial_i f} z) = 0$$
$$\Rightarrow z^T H_{\partial_i f} z = 0$$
$$\Rightarrow H_{\partial_i f} z = 0$$

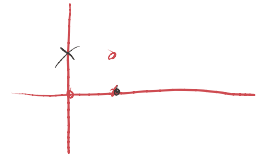
□

Week 6

C-Rayleigh = nonnegative coeff. 对于 $\alpha, w \in \mathbb{R}^n_{\geq 0}$

$$\partial^\alpha f(w)\, \partial^{\alpha+e_i+e_j} f(w) \leq C\, \partial^{\alpha+e_i} f(w)\, \partial^{\alpha+e_j} f(w)$$

* 反次m问题:



$$(1+x+xy) \leq C(x+xy)$$

let $x=y=0$.

Good: homogeneous. f C-Rayleigh $\Rightarrow$ supp(f) log-convex.

prop: if f C-Rayleigh, then

① $\partial_i f$ ← i 对 i 微分再删

② $f|_{i}$ 删掉 i 轴.

③ $f(a_1 w_1, \cdots, a_n w_n)$ → 换

④ $f(w_1+a_1, \cdots, w_n+a_n)$ → 把 $(a_i)_i \in \mathbb{R}^n_{\geq 0}$ 中间的点都看见

is C-Rayleigh.

Matroid $2^E = \{0,1\}^n$

① nonempty

② exchange property for $\mathcal{B}$.

$\alpha, \beta \in \mathcal{B}$, $i \in \alpha \setminus \beta$

$\Rightarrow \exists j \in \beta \setminus \alpha$ s.t. $(\alpha \setminus \{i\}) \cup \{j\} \in \mathcal{B}$.

① $\phi \in \mathcal{I}$.

② $\beta \in \mathcal{I}$, $\alpha \subseteq \beta \Rightarrow \alpha \in \mathcal{I}$

③ Augmentation property for $\mathcal{I}$.

$\alpha, \beta \in \mathcal{I}$, $|\alpha| < |\beta|$

$\Rightarrow \exists i \in \beta \setminus \alpha$, $\alpha \cup \{i\} \in \mathcal{I}$.

Convexity $\mathbb{N}^n$

① nonempty.

② $\mathcal{M}$-convex.

$\alpha, \beta \in J$, $\alpha_i > \beta_i$

$\Rightarrow \exists j$, $\alpha_j < \beta_j$ s.t. $\alpha - e_i + e_j \in J$.

① $0 \in J^\#$

② $\beta \in J^\#$, $\alpha \le \beta \Rightarrow \alpha \in J^\#$

($\alpha \le \beta \Leftrightarrow \alpha_i \in \beta_i, \forall i$)

③

$\alpha, \beta \in J^\#$, $|\alpha|_1 < |\beta|_1$

$\Rightarrow \exists i$, $\alpha_i < \beta_i$, $\alpha + e_i \in J^\#$.

$\mathcal{M}^\#$-convex.

$M^\#$-convex

① $0 \in J^\#$

② $\beta \in J^\#,\ \alpha \le \beta \Rightarrow \alpha \in J^\#$

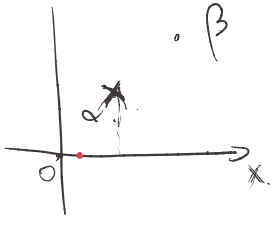
③ $\alpha, \beta \in J^\#,\ |\alpha|_1 < |\beta|_1$
$\Rightarrow \exists i,\ \alpha_i < \beta_i,\ \alpha + e_i \in J^\#$

thm: f c-Rayleigh, $\mathrm{supp}(f)$ ① $\Rightarrow$ ②③.
$\left(f(0) \neq 0 \Rightarrow \mathrm{supp}(f)\ M^\#\text{-convex}\right)$

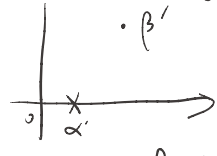
② Suppose.
Counterexample f. c-Rayleigh. $f(0) \neq 0$.
$\alpha \le \beta,\ \beta \in J^\#,\ \alpha \notin J^\#$
α minimal α minimal. $|\beta|_1$ minimal.

1° $\forall j,\ \beta_j \neq 0$ $\left(f|_j,\ \alpha \le \beta\right)$

2° $\alpha = e_i$ for some i
find $\alpha_j > 0$ { ① $e_j \notin \mathrm{supp}(f)$ $e_j \le \beta$
② $e_j \in \mathrm{supp}(f)$ $\partial_j f$.

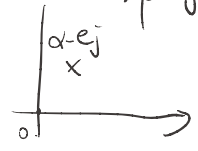


①.

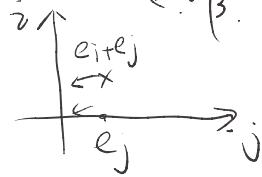


$\alpha' \le \beta'$

②



3° $e_j \in Supp(f) \Rightarrow e_i + e_j \in Supp(f)$,



$\Leftarrow$: β. (otherwise $\partial_j f$)

$\Rightarrow$. $f(0) \neq 0$. $\partial_i \partial_j f(0) \neq 0$.

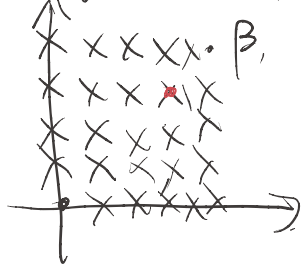
$\partial_i f(0) = 0 \Rightarrow f$ not c-Rayleigh.

$f(0)\,\partial_i\partial_j f(0) \leq c\,\partial_i f(0)\,\partial_j f(0)$

$\neq 0$ $\quad$ $= 0$.

$\Rightarrow e_j \notin Supp(f)$ $\forall j$

$\Rightarrow$ $\forall\, 0 \neq \gamma \leq \beta \Rightarrow \gamma \notin Supp(f)$.



(otherwise $\exists (e_j \leq \gamma)$

Counterexample)

$\forall l, k$, $w=(\xi, \cdots, \xi)$

$f = \sim + \sim\xi^{|\beta|} +$ 高次

$\partial_l \partial_k f = \sim \xi^{|\beta|-2} +$ 高次.

$\partial_l f = \sim \xi^{|\beta|-1} +$ 高次

$\partial_k f = \sim \xi^{|\beta|-1} +$ 高次

C-Rayleigh $\Rightarrow$

$\sim \xi^{|\beta|-2} +$ 高次 $\leq C \xi^{2|\beta|-2} +$ 高次.

$\Rightarrow$ Contradiction.

③ $\alpha, \beta \in J^{\#}$, $|\alpha| < |\beta|$, $\Rightarrow \exists i$, $\alpha_i < \beta_i$, $\alpha + e_i \in J^{\#}$

Counterexample f, α, β minimal α, n. $|\alpha|$

$\forall j$, $\alpha_j < \beta_j \Rightarrow \alpha + e_j \notin \mathrm{supp}(f)$

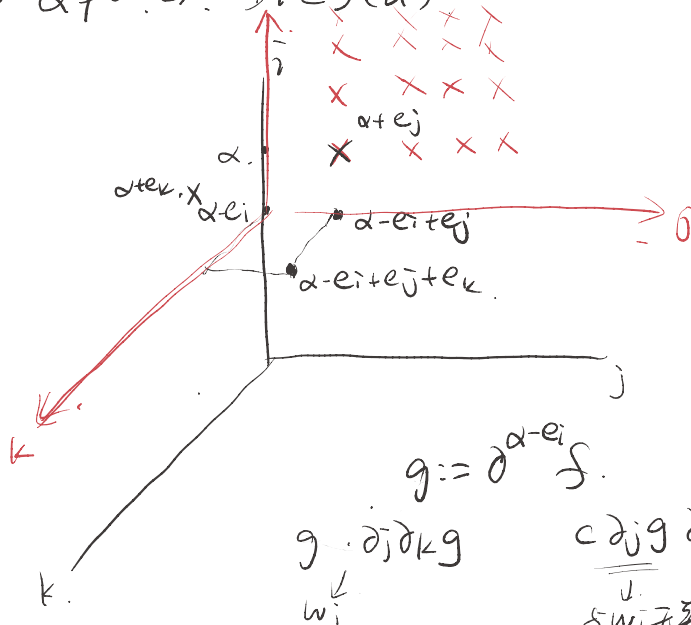
$S(\gamma) = \{i \mid \gamma_i > 0\}$.

1° $S(\alpha) \cap S(\beta) = \phi$ otherwise $j \in S(\alpha) \cap S(\beta)$.

$\partial_j f$ counter --

2° $S(\alpha) \cup S(\beta) = [n]$ otherwise $f \backslash j$

② $\Rightarrow \alpha \neq 0 \Rightarrow \exists i \in \mathrm{Supp}(\alpha)$



$g := \partial^{\alpha - e_i} f$.

$g \cdot \partial_j \partial_k g$ $\quad$ $c\, \partial_j g\, \partial_k g$

w_i ↓ $\quad$ ↓ $\sum w_i$ 系.

prop: $J^\natural$ $M^\natural$-convex. $m := \max_{\alpha \in J^\natural} |\alpha|_1$

$\Rightarrow J := \{\alpha \in J^\natural \mid |\alpha|_1 = m\}$ M-convex.

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f c-Rayleigh $\Rightarrow g(w) := f(w + \mathbf{1}_n)$ c-Rayleigh

$\Rightarrow \mathrm{supp}(g)$ $M^\natural$-convex

$\Rightarrow \mathrm{supp}(f)$ M-convex.

[Multi-affine poly f]

dfn : strongly Rayleigh multi-affine f.

$$f(w)\,\partial_i\partial_j f(w) \le \partial_i f(w)\,\partial_j f(w)$$

prop : M.A. strongly $\Leftrightarrow$ 1-Rayleigh

$$f(w) = w_k \partial_k f(w) + A.$$

$$(w_k\partial_k f + \sim)(w_k \partial_i\partial_j \partial_k f + \sim)$$

$$\le (w_k \partial_i \partial_k f + \sim)(w_k \partial_j \partial_k f + \sim)$$

$$(\partial_k f)(\partial_i\partial_j\partial_k f) \le (\partial_i\partial_k f)(\partial_j\partial_k f)$$

thm: M.A. strongly-Rayleigh $\Leftrightarrow$ stable.

cor: of $f \in H_n^d$ M.A stable $\Rightarrow$ Supp(f) M-convex

$\Rightarrow$ Supp(f) Matroid.